

CSCE 748 COMPUTATIONAL PHOTOGRAPHY · MODULE 04 · LECTURE NOTES

Tone Mapping

Showing a million to one on a display that does two hundred.

MODULE OUTCOMES

- Explain why tone mapping is underdetermined.
- Implement a global tone mapping operator.
- Implement a local operator and explain halos.
- Explain gradient-domain tone mapping.
- Evaluate tone mapping honestly.

1 Why there is no correct answer

tone mapping is underdetermined, and therefore a choice

Twenty stops of scene range must be mapped onto roughly eight stops of display range. Information will be lost. The only question is which information, and that question has no answer derivable from the data.

There is no objectively correct mapping, because the goal is not fidelity to the measured radiance — the display is physically incapable of reproducing it — but fidelity to the *impression* a viewer would have had of the original scene.

Which is a perceptual and ultimately aesthetic criterion, not a numerical one. Two operators can both be defensible and produce visibly different images.

So tone mapping operators are proposals rather than solutions, evaluated by how their output looks to people, and the literature is correspondingly unsettled after thirty years. **This is the same honest position CSCE 647's Module 13 reached from the rendering side**, and the two courses agree because it is a property of the problem rather than of either field.

	Global operators	Local operators
Mapping	One function applied identically to every pixel.	The mapping depends on the pixel's neighbourhood.
Consequence	The same input value always produces the same output value. Order is preserved everywhere.	A dark region is brightened and a bright region compressed, independently — so two pixels of equal radiance can map to different outputs.

	Global operators	Local operators
Local contrast	Necessarily reduced where the curve is flat — which is wherever it is doing the most compression.	Preserved throughout , because only the large-scale component is compressed.
Artefacts	None . Cannot produce halos or reversals.	Halos at strong edges (§3), and results can look unnatural or 'HDR' in the pejorative sense.
Examples	Gamma, Reinhard global, filmic curves, ACES.	Durand–Dorsey bilateral, Fattal gradient-domain, exposure fusion.

2 Global operators

$$L_{\text{scaled}} = (a / L_{\text{key}}) \cdot L \quad L_{\text{display}} = L_{\text{scaled}} / (1 + L_{\text{scaled}})$$

Reinhard's operator. The first step scales by a chosen *key* value a divided by the **log-average luminance** of the image, which sets the overall exposure automatically — the log average is used because perception of brightness is roughly logarithmic, so it corresponds to the subjective mid-tone. The second step applies a compressive curve mapping $[0, \infty)$ to $[0, 1)$, so **nothing ever clips**.

$$\text{with a white point: } L_d = L(1 + L/L_{\text{white}}^2) / (1 + L)$$

The extension allows values above a chosen white point to burn out deliberately rather than being compressed asymptotically. **This frequently looks better** — a photograph in which the sun is white is more natural than one in which it has been compressed into a grey disc, and viewers expect some highlights to clip.

OPERATE ON LUMINANCE, NOT ON THE COLOUR CHANNELS

Applying a compressive curve independently to R, G, and B compresses the differences between them, because the curve's slope falls as values rise.

So saturated colours desaturate as they brighten, and bright regions drift toward white. A saturated red highlight becomes pink and then white, which is not what the measurement said.

Instead: compute luminance, apply the tone curve to the luminance alone, and scale all three colour channels by the ratio of output luminance to input luminance. The ratios between channels — which is to say the hue and saturation — are then preserved exactly.

Then apply a saturation correction, typically a power on the chroma. This is necessary because the Hunt effect means perceived colourfulness falls with luminance — so an image whose luminance has been compressed looks less colourful even when the ratios are mathematically preserved, and a modest saturation boost restores the intended appearance.

3 Local operators and halos

HALOS COME FROM A BASE LAYER THAT CROSSES EDGES

Local operators decompose the image into a 'base' layer carrying the large-scale brightness variation and a 'detail' layer carrying local contrast, compress only the base, and recombine. The intuition is sound: the large-scale range is what exceeds the display, and local texture does not need compressing.

If the base layer is produced by a Gaussian blur, it smears across strong edges. Near a boundary between a bright window and a dark wall, the blurred base is a mixture of both — too dark on the window side and too bright on the wall side.

So the compression applied is wrong on both sides of the edge, and in opposite directions. The result is a bright halo along the dark side of the boundary and a dark halo along the bright side — the characteristic and immediately recognisable artefact of bad local tone mapping.

The fix is an edge-aware base layer (Module 02): a bilateral or guided filter, which does not average across strong edges and therefore produces a base that is correct on both sides. **This application is a large part of why the bilateral filter entered graphics at all.**

```
1. L      = luminance of the HDR image
2. log_L  = log(L)
3. base   = bilateral_filter(log_L)      # EDGE-AWARE, not Gaussian
   detail = log_L - base
4. base_c = (base - max(base)) * compression_factor
5. log_out = base_c + detail             # detail untouched
   L_out  = exp(log_out)
6. RGB_out = (RGB_in / L) * L_out ; then a saturation power
```

Compress the base, keep the detail. That structure recurs in nearly every local operator, and the differences between them are largely differences in how the decomposition is computed.

GRADIENT-DOMAIN TONE MAPPING

Fattal's reformulation: the thing that exceeds the display's range is not the absolute values but the large *gradients* — the big transitions between bright and dark regions. Texture consists of small gradients and is perfectly displayable.

So attenuate large gradients and leave small ones alone. A spatially varying attenuation function, larger where the gradient magnitude is larger, compresses the big transitions while preserving fine detail exactly.

Then integrate the modified gradient field back into an image. The modified field is not generally a valid gradient field of any image, so the integration is a least-squares problem — **which is a Poisson equation**, and is solved exactly as in Module 08.

It handles very large ranges gracefully and produces results without the halos of naive base/detail methods. **And it is the same machinery as seamless cloning**, which is why Module 08 follows immediately and treats the gradient domain as a topic in its own right.

4 Evaluating tone mapping honestly

- **There is no ground truth image to compare against.** The display cannot show the radiance map, so there is no reference rendering of it that would be correct. This rules out the RMSE-style evaluation that CSCE 647 relied on throughout.
- **So comparisons are necessarily perceptual** — images shown side by side, under controlled viewing conditions, to people who are asked to judge. The methodology of CSCE 650 Module 13 applies here.
- **Report the operator and its parameters.** A great deal of published comparison between imaging systems is, on inspection, a comparison between their default tone curves. **If two images are tone mapped differently, nothing else about them is being compared.**
- **Show what was lost.** Mark the clipped regions, show the ranges that were compressed, and present the pleasing result alongside the honest accounting rather than instead of it.
- **And distinguish 'accurate' from 'preferred'.** These are measurably different and have been studied: in forced-choice experiments people frequently prefer operators that are demonstrably less faithful to the original scene. Both are legitimate goals and they are not the same goal, and a paper claiming one while measuring the other is a common failure.

HDR DISPLAYS CHANGE THE QUESTION WITHOUT REMOVING IT

A display capable of a thousand nits or more with genuinely deep blacks requires far less compression — perhaps twelve stops rather than eight, which is a substantial fraction of what the sensor captured.

So tone mapping becomes a smaller intervention, and the displayed image can sit much closer to the measurement. The operator's choices matter less because it is choosing among fewer losses.

But it does not disappear. Real scenes still exceed any display by orders of magnitude, and the mapping now has to target a *range* of display capabilities rather than one assumed standard — the same content may be shown on a phone in sunlight and a reference monitor in a dark room.

Which is why HDR delivery standards carry metadata describing the content's actual range and intended appearance, letting each display perform the final mapping for its own capabilities. **This defers the decision to the last possible moment**, which is exactly what CSCE 647's Module 13 recommended for the same reason.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Reinhard et al. — Photographic Tone Reproduction for Digital Images (free)	The global operator of §2, including the photographic analogy that motivates it.

Resource	What it covers for this module
Durand & Dorsey — Fast Bilateral Filtering for the Display of HDR Images (free)	The bilateral tone mapping of §3, and one of the papers that brought the bilateral filter into graphics.
Fattal, Lischinski & Werman — Gradient Domain High Dynamic Range Compression (free)	The gradient-domain method, and a natural lead-in to Module 08.
Mantiuk et al. — tone mapping evaluation work (free)	Perceptual evaluation of tone mapping, including the accurate-versus-preferred distinction in §4.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Implement Reinhard's global operator with and without a white point. Compare on a scene with a visible light source.
- 2 Apply a tone curve per channel and to luminance only. Photograph something saturated and compare the desaturation.
- 3 Implement a saturation correction and find the exponent you prefer. Report it and note that it is a preference.
- 4 Implement base/detail tone mapping with a Gaussian base layer and **produce halos deliberately** on a high-contrast edge.
- 5 Replace the Gaussian with a bilateral filter and show the halos disappear.
- 6 Implement gradient-domain tone mapping, including the Poisson solve. Compare against the bilateral method on the same radiance map.
- 7 Tone map one radiance map with four operators and present them without labels to six people. Ask which is most accurate and which they prefer, separately.
- 8 Produce a figure showing, for one tone mapped image, which regions were clipped and how much each range was compressed.
- 9 If you have access to an HDR display, show the same radiance map on it and on an SDR display and describe the difference in what the operator had to do.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Why is tone mapping underdetermined, and what kind of criterion does that make it?

- 2 Compare global and local operators on five axes.
- 3 Write Reinhard's operator and say what the log-average luminance is for.
- 4 Why apply the curve to luminance rather than per channel, and why correct saturation afterwards?
- 5 Explain halos precisely, and give the fix.
- 6 Describe base/detail tone mapping in five steps.
- 7 What does gradient-domain tone mapping attenuate, and how is the result integrated?
- 8 Give five rules for evaluating tone mapping honestly.
- 9 What do HDR displays change, and what do they not?